

CONTAINERS OVER $\mathbf{Vec}$: THE BIPRODUCT COLLAPSE AND THE EXTENSIVITY CRUX

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ABSTRACT. We work out how much of the theory of containers survives the base change $\mathbf{Set} \rightsquigarrow \mathbf{Vec}$. A *linear container* is an object $(S, (P_s))$ of $\mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}})$ — a set of shapes, each carrying a vector space of positions — and its extension is the endofunctor $\llbracket S, P \rrbracket W = \bigoplus_s \mathbf{Vec}(P_s, W)$ on $\mathbf{Vec}$. The whole story is governed by a single fact: $\mathbf{Set}$ is extensive and $\mathbf{Vec}$ is not. In $\mathbf{Set}$ the coproduct is a disjoint union, and this is exactly what lets one read the shape set off the functor and makes the extension functor fully faithful. In $\mathbf{Vec}$ the coproduct is a biproduct, and the same two properties fail *together*: shapes become invisible (a finite linear container is Id^N , determined by one number N), and the extension functor stops being full (natural transformations may mix shapes; container morphisms may not). The gap is the single inclusion $\coprod \subsetneq \bigoplus$. The same failure demotes the comonoid theory — the $\triangleleft$ -comonoids of $\mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}})$ are only *families of $\mathbb{k}$ -algebras*, not algebroids — but here we can say exactly what went wrong and fix it: the correct home for a linear directed container is the matrix bicategory $\mathbf{Mat}(\mathbf{Vec})$, one hom-space per ordered pair of objects, whose comonoids *are* algebroids (Bénabou), and the single-index theory is its diagonal degeneration. The repair is to move the coproduct from the shape index to the composition index. Read for applications, the same biproduct is a feature: modelling a prompt as a one-shape linear container, “handle either prompt A or B ” is their coproduct, whose reply-space is the biproduct $A \oplus B$ — so “answer either” is the same object as “ready for both”. This is an orientation document for the grant’s “containers over other bases” front; the objects here are the additive corner of the strict polynomial functors — equivalently the degree-one slice of the analytic (Schur functor) lifting of containers to $\mathbf{Vec}$, where the Young-diagram classification degenerates to the single number N — and we are scrupulous throughout about what that literature owns and what the container framing adds.

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Date: 18 August 2026 (rev. 21 August 2026) — internal expository note.

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1. THE QUESTION, AND THE ONE-LINE ANSWER

The reader knows the **Set** container story. A *container* is a pair (S, P) with S a set of *shapes* and $P: S \rightarrow \mathbf{Set}$ a family of *position sets*; its extension is the endofunctor

$$\llbracket S, P \rrbracket X = \sum_{s \in S} \mathbf{Set}(P_s, X) = \sum_{s \in S} X^{P_s} \quad (\mathbf{Set} \rightarrow \mathbf{Set}).$$

Two facts make this theory rigid and useful. First, the shape set is *readable off the functor*: $\llbracket S, P \rrbracket(1) = \sum_s 1^{P_s} = S$, so evaluation at the terminal object returns the shapes. Second, the extension $\llbracket - \rrbracket: \mathbf{Fam}(\mathbf{Set}^{\text{op}}) \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is *fully faithful*, with essential image the polynomial (familiarily representable) functors; the data (S, P) can be reconstructed from F . Together these say: containers *are* their extensions, and the extensions wear their combinatorics on their sleeve.

Neil's steer opens a new front: keep the container idea but change the base, so that the *answer space is a vector space*. Fix a field $\mathbb{k}$ and let the positions be $\mathbb{k}$ -vector spaces and the extension land in **Vec**. Concretely we

study the single functor

$$\llbracket - \rrbracket : \mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}}) \longrightarrow [\mathbf{Vec}, \mathbf{Vec}], \quad \llbracket S, P \rrbracket W = \bigoplus_{s \in S} \mathbf{Vec}(P_s, W),$$

and ask the base-change question: *which parts of the $\mathbf{Set}$ theory carry over, and which break?*

The one-line answer. Everything is controlled by one difference between the bases:

$\mathbf{Set}$ is extensive — its coproduct is disjoint union — and $\mathbf{Vec}$ is not — its coproduct is a biproduct.

The two rigidity facts above both rest on extensivity, and so both fail, and they fail as *one* phenomenon. Precisely:

- (a) **Shapes become invisible (objects).** The terminal object of $\mathbf{Vec}$ is the *zero* object, so $\llbracket S, P \rrbracket(0) = 0$: there is no “ $F(1) = S$ ”. Worse, for finite data $\llbracket S, P \rrbracket \cong \mathrm{Id}^N$ with $N = \sum_s \dim P_s$, so the functor remembers only the total dimension N — not the shapes, not even the partition of N into shapes (Theorem 5.1).
- (b) **The extension stops being full (morphisms).** Natural transformations between extensions are indexed by the *biproduct* $\bigoplus_t$, while container morphisms are indexed by the *disjoint union* $\coprod_t$; the comparison is the inclusion $\coprod_t \hookrightarrow \bigoplus_t$, which misses every transformation that mixes two shapes (Theorem 6.3).

Both bullets are the single inclusion $\coprod \subsetneq \bigoplus$: in $\mathbf{Set}$ these two coproducts coincide (extensivity), in $\mathbf{Vec}$ they do not. That is the whole paper.

What is new, and what is not. The *objects* here — functors $W \mapsto \bigoplus_s \mathbf{Vec}(P_s, W)$ — are the additive (degree-one) corner of the strict polynomial functors of Friedlander–Suslin, a highly developed theory; the analytic/Day side is the theory of vector species and twisted commutative algebras; “coproduct of representables” and its extensivity hypothesis belong to Diers’ familial representability; “algebroid” is Mitchell’s. None of these objects is ours. What the container framing *adds* is the assembly: it introduces a *shape index* on top of these functors and then discovers, through the computation of this note, that the shape index is not an invariant of the functor — and that its collapse (on objects) and the failure of fullness (on morphisms) are the same event, the failure of extensivity. See §12 for the honest ledger.

Roadmap. §2 is a short recap of the base $\mathbf{Vec}$ itself — what it has (a zero object, biproducts, self-enrichment) and what it lacks that $\mathbf{Set}$ has (extensivity above all) — for the reader fluent in categories but new to the linear instance. §3 gives the definitions ($\mathbf{Set}$ and $\mathbf{Vec}$ side by side). §4 proves the three-lemma toolbox (a linear co-Yoneda lemma; $h_P = \mathbf{Vec}(P, -)$ is additive and fully faithful; finite positions distribute over $\bigoplus$). §5 is the biproduct collapse. §6 is the hom formula and the extensivity crux. §7 computes composition and probes comonoids (the algebroid guess, honestly assessed). §8

then resolves it: the correct home for a linear directed container is the matrix bicategory $\mathbf{Mat}(\mathbf{Vec})$, whose (co)monoids are genuine algebroids, and the single-index theory is exactly its diagonal degeneration. §9 is the application face: handling either of two prompts is the coproduct of their linear containers, and the biproduct collapse makes “answer either” the same object as “ready for both”. §10 is the carry-over table. §11 sets our objects against the analytic (Schur functor) lifting and locates them as its degree-one corner. §12 is the prior-art ledger, and §13 the open questions.

2. THE STRUCTURE OF $\mathbf{Vec}$, FOR THE CONTAINER-MINDED

Before the definitions, a short recap of the base itself. The reader here is fluent in categories but may simply not have worked with the linear-algebra instance; the whole note turns on a handful of structural facts about $\mathbf{Vec}$, and it is worth having them in one place, side by side with $\mathbf{Set}$. Nothing in this section is proved — each fact is standard, and each is a signpost to the section where it does real work.

What $\mathbf{Vec}$ has.

A zero object.:

$\mathbf{Vec}$ is *pointed*: the space 0 is at once terminal and initial. There is a unique map $V \rightarrow 0$ and a unique map $0 \rightarrow V$, and they are the same object. This is the first thing that breaks the $\mathbf{Set}$ intuition, where terminal (1) and initial $(\emptyset)$ are wildly different sets.

Finite biproducts.:

For finitely many spaces the coproduct and the product *coincide*: $V_1 \oplus \cdots \oplus V_n$ carries both the injections $V_i \hookrightarrow \bigoplus$ (making it a coproduct) and the projections $\bigoplus \rightarrow V_i$ (making it a product), compatibly. This shared object is the *biproduct*. It is the structural heart of the note: every collapse below is this one coincidence, viewed through a different lens, and §9 reads it as a *feature*.

Kernels and cokernels.:

$\mathbf{Vec}$ is *abelian*: every map has a kernel and a cokernel, and every mono/epi is normal. Idempotents split; the endomorphism ring of a functor governs its decomposition into indecomposables (Krull–Schmidt–Azumaya, used in Theorem 5.1).

Self-enrichment.:

$\mathbf{Vec}$ is enriched over itself: each $\mathrm{hom} \mathbf{Vec}(V, W)$ is again a *vector space*, and composition is bilinear. This is what makes “linear container” more than a slogan — the extension functor is a $\mathbf{Vec}$ -functor, and the linear co-Yoneda lemma (Lemma 4.1), the engine of every computation below, is exactly the enriched Yoneda lemma read in $\mathbf{Vec}$.

What $\mathbf{Vec}$ lacks that $\mathbf{Set}$ has.

Extensivity.:

This is the one that matters. $\mathbf{Set}$ is *extensive*: its coproduct is a genuine disjoint union, and a map into $A \sqcup B$ is a *partition* of the domain into

the part landing in A and the part landing in B (Carboni–Lack–Walters). $\mathbf{Vec}$ is *not* extensive: its finite coproduct is the biproduct, and a map into $A \oplus B$ is a *pair* of maps (into A and into B) that may both be nonzero. The comparison is the strict inclusion

$$\amalg \subsetneq \bigoplus,$$

and every “fails” in this note is one avatar of it (Theorems 5.1, 6.3).

Disjoint, complemented injections.:

In $\mathbf{Set}$ the coproduct injections are disjoint ($A \cap B = \emptyset$ inside $A \sqcup B$) and each summand is complemented. In $\mathbf{Vec}$ the injections $A, B \hookrightarrow A \oplus B$ overlap in every vector $a + b$ with $a, b \neq 0$; there is no “which summand did this element come from?”. That missing partition is precisely the information a shape function records and a biproduct destroys.

A subobject classifier.:

$\mathbf{Set}$ is a topos; $\mathbf{Vec}$ is not. There is no Ω , so “subset” is not representable and the topos-theoretic tools that support the $\mathbf{Set}$ container calculus are unavailable. (We do not use them; the note leans on additivity and Krull–Schmidt instead.)

A shape-detecting terminal object.:

In $\mathbf{Set}$ the terminal object 1 is a *separator* that sees shapes: $\llbracket S, P \rrbracket(1) = S$. In $\mathbf{Vec}$ the terminal object is 0 , and $\llbracket S, P \rrbracket(0) = 0$ sees nothing. The slogan “shapes = points of $F(1)$ ” is a fact about $\mathbf{Set}$, not about containers, and it does not survive the base change (§5).

The punchline of the recap. Read the two lists against each other and a single line does all the work. Everything $\mathbf{Vec}$ *has* that we use — biproducts, self-enrichment, Krull–Schmidt — and the one thing it *lacks* that we miss — extensivity — are two faces of the same coin: the finite coproduct is a biproduct. Where $\mathbf{Set}$ keeps its summands disjoint, $\mathbf{Vec}$ merges them, and $\amalg \subsetneq \bigoplus$ is the record of that merge. The rest of the note is the systematic consequence of this one difference, and §9 shows that the “merge” is not only a liability for a representation theorem but, read on the coproduct of two prompts, exactly the structure an application wants.

3. DEFINITIONS

Throughout, $\mathbb{k}$ is a fixed field, $\mathbf{Vec} = \mathbf{Vec}_{\mathbb{k}}$ is the category of all $\mathbb{k}$ -vector spaces, $\mathbf{Vec}(V, W)$ is the hom *vector space*, and $[\mathbf{Vec}, \mathbf{Vec}]$ is the category of $\mathbb{k}$ -linear (= additive) endofunctors and natural transformations. Every functor $F: \mathbf{Vec} \rightarrow \mathbf{Vec}$ below is a $\mathbf{Vec}$ -functor: its action on homs $\mathbf{Vec}(V, W) \rightarrow \mathbf{Vec}(FV, FW)$ is $\mathbb{k}$ -linear.

Definition 3.1 (linear container). A *linear container* is an object of $\mathbf{LinCont} := \mathbf{Fam}(\mathbf{Vec}^{\text{op}})$: a pair $(S, (P_s)_{s \in S})$ with S a set (the *shapes*) and each $P_s \in \mathbf{Vec}$ (the *position space* of shape s). A morphism $(S, (P_s)) \rightarrow (T, (Q_t))$ is a pair $(f, (\varphi_s))$ where $f: S \rightarrow T$ is a function

and, for each $s \in S$, $\varphi_s: Q_{f(s)} \rightarrow P_s$ is a linear map — contravariant on positions, exactly as for **Set** containers. Write $n_s := \dim P_s$.

Compare the **Set** definition: there P_s is a *set* and $\varphi_s: Q_{f(s)} \rightarrow P_s$ a *function*. Everything is the same except the base of the position fibres, $\mathbf{Set}^{\text{op}} \rightsquigarrow \mathbf{Vec}^{\text{op}}$.

Example 3.2. $(\{*\}, \mathbb{k}^2)$ is the one-shape container whose single shape has a two-dimensional position space. $(\{*_1, *_2\}, (\mathbb{k}, \mathbb{k}))$ is the two-shape container, each shape one-dimensional. These are *different* objects of $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ (different shape sets). Watch them: they are the running counterexample of the whole note.

Definition 3.3 (extension). The *extension* of (S, P) is the functor $\llbracket S, P \rrbracket: \mathbf{Vec} \rightarrow \mathbf{Vec}$,

$$\llbracket S, P \rrbracket W := \bigoplus_{s \in S} \mathbf{Vec}(P_s, W).$$

Choosing a basis of P_s gives $\mathbf{Vec}(P_s, W) \cong W^{n_s}$ naturally in W , so $\llbracket S, P \rrbracket W \cong \bigoplus_{s \in S} W^{n_s}$, a “direct sum of powers”. On a morphism $(f, (\varphi_s))$ the natural transformation acts by $(g_s)_s \mapsto (g_{f(s)} \circ \varphi_s)_s$. This makes $\llbracket - \rrbracket: \mathbf{Fam}(\mathbf{Vec}^{\text{op}}) \rightarrow [\mathbf{Vec}, \mathbf{Vec}]$ a functor.

Definition 3.4 (corepresentable). For $P \in \mathbf{Vec}$ write $h_P := \mathbf{Vec}(P, -): \mathbf{Vec} \rightarrow \mathbf{Vec}$, the *corepresentable* at P . Then $\llbracket S, P \rrbracket = \bigoplus_{s \in S} h_{P_s}$, and $h_{\mathbb{k}} = \mathbf{Vec}(\mathbb{k}, -) \cong \text{Id}$.

So a linear container’s extension is a coproduct of corepresentables, and the unit corepresentable is the identity functor. This is the analog of “a **Set** container is a coproduct of representables X^{P_s} ”, with $X^{P_s} = \mathbf{Set}(P_s, X)$ the represented functor at the set P_s .

A non-example. The functor $W \mapsto W \otimes_{\mathbb{k}} W$ (tensor square) is a **Vec**-endofunctor but is *not* the extension of any linear container: it is homogeneous of degree 2 (it scales like λ^2 under $W \xrightarrow{\lambda} W$), whereas every $\llbracket S, P \rrbracket$ is additive (degree 1). Linear containers see only the degree-one world; the higher-degree strict polynomial functors (symmetric, exterior, tensor powers) live outside $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$. This is the first hint that “linear container” means “functor built from Id under $\bigoplus$ ” — more precisely, as Remark 5.2 sharpens, a *cocontinuous* **Vec**-endofunctor with finite-dimensional value on $\mathbb{k}$ — which the collapse theorem will make exact.

4. THE THREE-LEMMA TOOLBOX

Everything downstream is assembled from three short lemmas: a linear co-Yoneda lemma (how maps out of a corepresentable are pinned down by one element), the additivity and full faithfulness of h , and the fact that finite-dimensional positions let hom commute with $\bigoplus$. Prove these once and the rest is bookkeeping.

Lemma 4.1 (linear co-Yoneda). *For any $\mathbf{Vec}$ -functor $G: \mathbf{Vec} \rightarrow \mathbf{Vec}$ and any $P \in \mathbf{Vec}$ there is an isomorphism of vector spaces $\mathrm{Nat}(h_P, G) \cong G(P)$, natural in G and P . In particular $\mathrm{Nat}(\mathrm{Id}, G) \cong G(\mathbb{k})$ and $\mathrm{Nat}(h_P, h_Q) \cong \mathrm{Vec}(Q, P)$.*

Proof. Given $\alpha: h_P \Rightarrow G$, set $\xi := \alpha_P(\mathrm{id}_P) \in G(P)$. For any $\varphi: P \rightarrow W$, naturality of α along the postcomposition map $\varphi_*: \mathrm{Vec}(P, P) \rightarrow \mathrm{Vec}(P, W)$ gives

$$\alpha_W(\varphi) = \alpha_W(\varphi_* \mathrm{id}_P) = G(\varphi)(\alpha_P \mathrm{id}_P) = G(\varphi)(\xi),$$

so α is determined by ξ . Conversely, for $\xi \in G(P)$ define $\alpha_W(\varphi) := G(\varphi)(\xi)$; this is linear in φ because G is a $\mathbf{Vec}$ -functor (so $\varphi \mapsto G(\varphi)$ is linear) followed by evaluation at ξ , and natural in W by functoriality of G . The two constructions are mutually inverse and $\mathbb{k}$ -linear in ξ , so $\mathrm{Nat}(h_P, G) \cong G(P)$. For $G = h_Q$, $G(P) = \mathrm{Vec}(Q, P)$. $\square$

Remark 4.2. This is the enriched co-Yoneda lemma read in $\mathbf{Vec}$; the only wrinkle over the ordinary Yoneda argument is that we need $\varphi \mapsto \alpha_W(\varphi)$ to be *linear*, which is exactly what “ G is a $\mathbf{Vec}$ -functor” buys us.

Lemma 4.3 (h is additive and fully faithful). *The assignment $h: \mathbf{Vec}^{\mathrm{op}} \rightarrow [\mathbf{Vec}, \mathbf{Vec}]$, $P \mapsto h_P$, is fully faithful and sends finite biproducts to biproducts: $h_{P \oplus Q} \cong h_P \oplus h_Q$. Consequently the only indecomposable corepresentable is $h_{\mathbb{k}} \cong \mathrm{Id}$, and $\mathrm{End}(\mathrm{Id}) = \mathbb{k}$.*

Proof. Full faithfulness is Lemma 4.1 with $G = h_Q$: $\mathrm{Nat}(h_P, h_Q) \cong \mathrm{Vec}(Q, P) = \mathbf{Vec}^{\mathrm{op}}(P, Q)$. Additivity: $h_{P \oplus Q}(W) = \mathbf{Vec}(P \oplus Q, W) \cong \mathbf{Vec}(P, W) \times \mathbf{Vec}(Q, W) = (h_P \oplus h_Q)(W)$, using that a finite product in $\mathbf{Vec}$ is a biproduct and that biproducts in $[\mathbf{Vec}, \mathbf{Vec}]$ are computed pointwise; naturality is clear. Since any P with $\dim P \geq 2$ splits as $\mathbb{k} \oplus (\dots)$ and h is additive, h_P decomposes unless $\dim P = 1$; and $h_0 = 0$. So the unique indecomposable corepresentable is $h_{\mathbb{k}}$. Finally $\mathrm{End}(\mathrm{Id}) = \mathrm{Nat}(\mathrm{Id}, \mathrm{Id}) \cong \mathrm{Id}(\mathbb{k}) = \mathbb{k}$ by Lemma 4.1. $\square$

Lemma 4.4 (finite positions distribute over $\bigoplus$). *If $\dim P < \infty$ then for any family $(V_t)_{t \in T}$, $\mathbf{Vec}(P, \bigoplus_t V_t) \cong \bigoplus_t \mathbf{Vec}(P, V_t)$, naturally. Equivalently, h_P preserves coproducts iff P is finite-dimensional.*

Proof. Let $e_1, \dots, e_n$ be a basis of P . A linear map $\varphi: P \rightarrow \bigoplus_t V_t$ is determined by the finite tuple $(\varphi e_1, \dots, \varphi e_n)$, and each φe_i has finite support in T ; hence $\mathrm{im} \varphi$ lies in a finite subsum $\bigoplus_{t \in F} V_t$. Therefore φ lies in $\bigoplus_t \mathbf{Vec}(P, V_t)$ (the finitely supported families): this is the inclusion $\supseteq$; the reverse inclusion is immediate. For P infinite-dimensional the identity $P \rightarrow P = \bigoplus_{i \in I} \mathbb{k}$ has infinite support, so it lies in $\prod_i \mathbb{k}$ but not $\bigoplus_i \mathbb{k}$, and the isomorphism fails. $\square$

The three lemmas already announce the theme. Lemma 4.3 says the only indecomposable brick is Id ; Lemma 4.4 says that with finite data the extension is a *coproduct* of copies of Id ; and once a coproduct of copies of Id

over a *finite* index becomes a biproduct, the shapes are gone. That is the next section.

5. PART 1 — THE BIPRODUCT COLLAPSE

5.1. Terminal recovery fails; the replacement. The terminal object of $\mathbf{Vec}$ is the zero object 0 (terminal = initial in an additive category). Hence

$$\llbracket S, P \rrbracket(0) = \bigoplus_{s \in S} \mathbf{Vec}(P_s, 0) = \bigoplus_s 0 = 0.$$

Teachbox: always ask what the terminal object is. The $\mathbf{Set}$ slogan “ $F(1) = S$ ” is not a theorem about containers; it is a theorem about $\mathbf{Set}$, whose terminal object 1 is a *separator* that detects shapes. Change the base to $\mathbf{Vec}$, whose terminal object is 0 , and the slogan evaporates: $\llbracket S, P \rrbracket(0) = 0$ tells you nothing. The lesson generalises: when you port a construction to a new base, the *first* thing to recompute is the terminal object and what it can see.

The natural evaluative replacement — evaluate at the monoidal unit $\mathbb{k}$ instead of at the terminal object — fares only slightly better:

$$\llbracket S, P \rrbracket(\mathbb{k}) = \bigoplus_s \mathbf{Vec}(P_s, \mathbb{k}) = \bigoplus_s P_s^*, \quad \dim = \sum_s n_s.$$

This records the *total* position dimension $N = \sum_s n_s$, and even that only as a bare vector space — with no partition into shapes. The honest replacement for “shapes = points of $F(1)$ ” is not evaluative but *homological*:

shapes over $\mathbf{Vec}$ = indecomposable direct summands of the functor F

— and the next theorem measures how badly even that differs from the naive shape set.

5.2. Finite collapse.

Theorem 5.1 (finite collapse). *Let S be finite and every $n_s = \dim P_s$ finite, and set $N := \sum_{s \in S} n_s = \dim(\bigoplus_s P_s)$. Then there is a natural isomorphism $\llbracket S, P \rrbracket \cong \mathrm{Id}^N$. Consequently:*

- (1) $\llbracket S, P \rrbracket \cong \llbracket S', P' \rrbracket$ as functors iff $N = N'$. A finite linear container is classified up to isomorphism of its extension by the single number N .
- (2) Id is indecomposable with $\mathrm{End}(\mathrm{Id}) = \mathbb{k}$, and $\mathrm{End}(\mathrm{Id}^N) = M_N(\mathbb{k})$ is semiperfect, so by the Krull–Schmidt–Azumaya theorem the decomposition $\llbracket S, P \rrbracket \cong \mathrm{Id}^{\oplus N}$ into indecomposables is unique up to isomorphism and permutation. The recovered invariant is exactly the multiplicity N ; the shape set S and the partition $N = \sum_s n_s$ are not recoverable.

Proof. Choosing a basis of each P_s , $\mathbf{Vec}(P_s, W) \cong W^{n_s}$ naturally in W . Since S is finite,

$$\llbracket S, P \rrbracket W = \bigoplus_{s \in S} \mathbf{Vec}(P_s, W) \cong \bigoplus_{s \in S} W^{n_s} = W^{\sum_s n_s} = \mathrm{Id}^N(W),$$

naturally in W ; this is the isomorphism. Part (1): $\mathrm{Id}^N \cong \mathrm{Id}^{N'}$ iff $N = N'$ (evaluate at $\mathbb{k}$). Part (2): Id is indecomposable because $\mathrm{End}(\mathrm{Id}) = \mathbb{k}$ has no idempotents $\neq 0, 1$; $[\mathbf{Vec}, \mathbf{Vec}]$ is additive and idempotent-complete (idempotents split pointwise, as $\mathbf{Vec}$ is), and $\mathrm{End}(\mathrm{Id}^N) = M_N(\mathbb{k})$ is Artinian hence semiperfect, so Krull–Schmidt–Azumaya applies and gives uniqueness of the multiplicity. $\square$

Remark 5.2 (additive is not enough — the hypothesis is *cocontinuous*). The collapse is a statement about *cocontinuous* functors, not merely additive ones, and the distinction is real. Additivity alone does not force Id^N : the corepresentable $h_P = \mathbf{Vec}(P, -)$ with $\dim P = \infty$ is additive but a *product* functor $\prod_I W$, not a coproduct of copies of Id (Lemma 4.4); and $W \mapsto W^{**}$ is additive yet not of the form Id^N . The correct hypothesis is *cocontinuity* (preservation of coproducts). By the *Eilenberg–Watts theorem*, a $\mathbf{Vec}$ -cocontinuous $\mathbb{k}$ -linear $F: \mathbf{Vec} \rightarrow \mathbf{Vec}$ is $F \cong F(\mathbb{k}) \otimes_{\mathbb{k}} - \cong \mathrm{Id}^{\oplus \dim F(\mathbb{k})}$; so $\dim F(\mathbb{k}) = N < \infty$ gives $F \cong \mathrm{Id}^N$, with the only indecomposable being Id . A *finite* linear container is cocontinuous — this is exactly Lemma 4.4 (finite positions distribute over $\bigoplus$) together with finiteness of S — which is why Theorem 5.1 applies. This is the intrinsic characterization made sharp in Corollary 6.4, and it is the single clean use of Eilenberg–Watts in this note.

Example 5.3 (the collapse in one picture). Return to the running pair. Both

$$(\{*\}, \mathbb{k}^2) \quad \text{and} \quad (\{*_1, *_2\}, (\mathbb{k}, \mathbb{k}))$$

have extension Id^2 , i.e. $W \mapsto W^2$, though they have different shape sets. Over $\mathbf{Set}$ the analogous containers $(\{*\}, 2)$ and $(\{*_1, *_2\}, (1, 1))$ have *distinct* extensions X^2 and $2 \cdot X$, and these are separated at $X = 1$: $1^2 = 1 \neq 2 = 2 \cdot 1$. The collapse over $\mathbf{Vec}$ is *precisely the loss of that separator* — there is no object of $\mathbf{Vec}$ at which W^2 and $W \sqcup W$ come apart, because $\mathbf{Vec}$ has no separating object that behaves like the one-point set.

5.3. Where content survives. The collapse used *both* finiteness of S (to turn $\bigoplus$ into a finite biproduct) *and* finite-dimensionality of the positions (Lemma 4.4). Drop either and content returns.

Infinite S . With all $P_s = \mathbb{k}$, $\llbracket S, P \rrbracket = \bigoplus_{s \in S} \mathrm{Id} = \mathrm{Id}^{\oplus |S|}$. For infinite S this is *not* representable and $\bigoplus_s \mathrm{Id} \neq \prod_s \mathrm{Id}$: the functor is genuinely new. Still, Krull–Schmidt–Azumaya recovers only the cardinal $|S|$ (the number of Id summands), not any finer shape structure.

Infinite-dimensional positions. By Lemma 4.4, h_P for $\dim P = \infty$ does *not* preserve coproducts: $h_P(W) = \mathbf{Vec}(P, W) \cong \prod_{i \in I} W$ is an *infinite product* functor, not a coproduct of copies of Id . Such h_P are not built from Id under $\bigoplus$, and by Lemma 4.3 distinct infinite-dimensional P give non-isomorphic h_P . Here P is recovered from a single representable.

The morphism layer.: Even where objects collapse, the category $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ and the natural-transformation spaces between extensions do not. That is Part 2, and it is where the extensivity crux lives.

6. PART 2 — THE HOM FORMULA AND THE EXTENSIVITY CRUX

The **Set** representation theorem says $\llbracket - \rrbracket : \mathbf{Fam}(\mathbf{Set}^{\text{op}}) \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is fully faithful, with essential image the polynomial functors, and (S, P) recovered from F as $S = \pi_0(\text{el } F)$ with positions the generic elements. We isolate which half survives over $\mathbf{Vec}$ and locate the failure in one symbol.

Theorem 6.1 (hom formula). *For linear containers (S, P) , (T, Q) ,*

$$\text{Nat}(\llbracket S, P \rrbracket, \llbracket T, Q \rrbracket) \cong \prod_{s \in S} \bigoplus_{t \in T} \mathbf{Vec}(Q_t, P_s).$$

Proof. $\llbracket S, P \rrbracket = \bigoplus_s h_{P_s}$ is a coproduct, so $\text{Nat}(\bigoplus_s h_{P_s}, G) \cong \prod_s \text{Nat}(h_{P_s}, G) \cong \prod_s G(P_s)$ by Lemma 4.1. With $G = \llbracket T, Q \rrbracket$, $G(P_s) = \bigoplus_t \mathbf{Vec}(Q_t, P_s)$. $\square$

Theorem 6.2 (container-hom). *The morphisms of $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ are*

$$\mathbf{Fam}(\mathbf{Vec}^{\text{op}})((S, P), (T, Q)) \cong \prod_{s \in S} \left(\coprod_{t \in T} \mathbf{Vec}(Q_t, P_s) \right),$$

where $\coprod$ is the coproduct in **Set** (disjoint union): a morphism assigns to each s a choice $t = f(s)$ together with a linear map in $\mathbf{Vec}(Q_{f(s)}, P_s)$.

Proof. Immediate from Definition 3.1: a morphism is $f : S \rightarrow T$ plus, for each s , an element $\varphi_s \in \mathbf{Vec}(Q_{f(s)}, P_s)$; packaging $(f(s), \varphi_s)$ as an element of the disjoint union $\coprod_t \mathbf{Vec}(Q_t, P_s)$ and ranging over s gives the product. $\square$

Put the two formulas side by side. They are identical except for one symbol:

$$\underbrace{\prod_s \coprod_t \mathbf{Vec}(Q_t, P_s)}_{\text{container morphisms}} \longrightarrow \underbrace{\prod_s \bigoplus_t \mathbf{Vec}(Q_t, P_s)}_{\text{natural transformations}}$$

The map between them is the extension functor on homs. It is, component-wise, the comparison from the coproduct-in-**Set** to the coproduct-in-**Vec**.

Theorem 6.3 (the extensivity crux). *The functor $\llbracket - \rrbracket$ induces, on each hom, the canonical map*

$$\prod_s \coprod_t \mathbf{Vec}(Q_t, P_s) \longrightarrow \prod_s \bigoplus_t \mathbf{Vec}(Q_t, P_s),$$

which is componentwise the inclusion of the disjoint union $\coprod_t$ into the biproduct $\bigoplus_t$, sending a choice (t, φ) to the element supported at t . Therefore:

- (1) **Not full.** As soon as some shape $s \in S$ admits two shapes $t \neq t' \in T$ with $\mathbf{Vec}(Q_t, P_s) \neq 0 \neq \mathbf{Vec}(Q_{t'}, P_s)$, there are natural transformations $\llbracket S, P \rrbracket \Rightarrow \llbracket T, Q \rrbracket$ outside the image of $\llbracket - \rrbracket$ — namely any whose s -component is a genuine linear combination $\varphi_t + \varphi_{t'}$ supported on two shapes. The cokernel of the inclusion measures exactly the cross-shape linear combinations that the biproduct permits and a shape function f forbids.
- (2) **Extensivity is the whole story.** Over $\mathbf{Set}$ the same computation gives $\mathbf{Nat}(\llbracket S, P \rrbracket, \llbracket T, Q \rrbracket) \cong \prod_s \bigoplus_t \mathbf{Set}(Q_t, P_s)$, where now $\bigoplus_t$ is the coproduct in $\mathbf{Set}$ — the disjoint union $\coprod_t$. Hence container-hom = $\mathbf{Nat}$ and $\llbracket - \rrbracket$ is fully faithful: the classical representation theorem. The single change under $\mathbf{Set} \rightsquigarrow \mathbf{Vec}$ is that the coproduct in the hom formula splits into two inequivalent notions, $\coprod_t \subsetneq \bigoplus_t$.

Proof. The description of the induced map is the naturality of the $\llbracket - \rrbracket$ -action against Theorems 6.1–6.2: a container morphism $(f, (\varphi_s))$ produces the natural transformation whose s -component is the composite through the single summand $t = f(s)$, i.e. the element of $\bigoplus_t \mathbf{Vec}(Q_t, P_s)$ supported at $f(s)$. The map $\coprod_t X_t \rightarrow \bigoplus_t X_t$ (choice $\mapsto$ supported element) is injective away from zero and misses every element with support of size ≥ 2 ; taking $X_t = \mathbf{Vec}(Q_t, P_s) \neq 0$ for two values of t exhibits a missed element, proving (1). For (2): in $\mathbf{Set}$, $\mathbf{Nat}(\llbracket S, P \rrbracket, \llbracket T, Q \rrbracket) \cong \prod_s \llbracket T, Q \rrbracket(P_s) = \prod_s \sum_t \mathbf{Set}(Q_t, P_s)$, and $\sum_t = \coprod_t$, so this equals $\mathbf{Fam}(\mathbf{Set}^{\text{op}})((S, P), (T, Q))$; the induced map is the identity and $\llbracket - \rrbracket$ is fully faithful. The sole difference over $\mathbf{Vec}$ is $\bigoplus_t \neq \coprod_t$. $\square$

$$\begin{array}{ccc} s & \xrightarrow{\varphi} & t \\ & & t' \end{array}$$

container morphism: *one* branch, $\coprod_t$

$$\begin{array}{ccc} s & \xrightarrow{\varphi_t} & t \\ & \searrow \varphi_{t'} & \\ & & t' \end{array}$$

natural transformation: a combination, $\bigoplus_t$

The left morphism lives in $\coprod_t$: it *chooses* the target shape $t = f(s)$. The right transformation lives in $\bigoplus_t$: it sends s into a *linear combination* $\varphi_t + \varphi_{t'}$ of two target shapes at once. In $\mathbf{Set}$ there is no such combination — a function must pick one branch — which is exactly why the two hom sets coincide there. In $\mathbf{Vec}$ the biproduct manufactures the combinations, and they are precisely the transformations no container morphism can hit.

Corollary 6.4 (finite recovery, sharp). *Restrict to finite containers with finite-dimensional positions. Then $\llbracket - \rrbracket$ recovers from $\llbracket S, P \rrbracket$:*

- the total position space $\bigoplus_{s \in S} P_s \in \mathbf{Vec}$ up to isomorphism (equivalently the number N), because $\bigoplus_s h_{P_s} \cong h_{\bigoplus_s P_s}$ (Lemma 4.3) and h is fully faithful, so the extension determines $\bigoplus_s P_s$ exactly;
- nothing about the partition of N into shapes.

Thus the honest representation statement over $\mathbf{Vec}$ is: F is a finite linear container $\iff F \cong \mathrm{Id}^N \iff F$ is $\mathbf{Vec}$ -cocontinuous with $F(\mathbb{k})$ finite-dimensional; and the data (S, P) is recovered only up to the biproduct ambiguity — a single object $\bigoplus_s P_s$ with no canonical shape decomposition.

Proof. The two bullets are Theorem 5.1 and Lemma 4.3. For the intrinsic characterization: any $\mathbf{Vec}$ -cocontinuous F with $F(\mathbb{k})$ finite-dimensional is $F \cong F(\mathbb{k}) \otimes -$ by the Eilenberg–Watts theorem, and $\dim F(\mathbb{k}) = N < \infty$ forces $F(\mathbb{k}) \otimes - \cong \mathrm{Id}^N$; conversely Id^N is manifestly of this form. $\square$

7. PART 3 — COMPOSITION AND COMONIDS

Set containers compose by substitution, and the composite has shapes the dependent sum $\coprod_{s \in S} T^{P_s}$ and positions the dependent pairs. Over $\mathbf{Vec}$ the dependent sum flattens.

Proposition 7.1 (composition of linear containers). *For linear containers with finite-dimensional positions,*

$$[[S, P]] \circ [[T, Q]] \cong [[S \times T, (P_s \otimes Q_t)_{(s,t)}]],$$

so on $\mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}})$ (finite-dimensional positions) composition is $(S, P) \triangleleft (T, Q) = (S \times T, (P_s \otimes Q_t))$, with unit $I = (\{*\}, \mathbb{k})$ (extension Id).

Proof. We compute the composite on an object W :

$$[[S, P]]([T, Q] W) = \bigoplus_s \mathbf{Vec}\left(P_s, \bigoplus_t \mathbf{Vec}(Q_t, W)\right).$$

Since P_s is finite-dimensional, Lemma 4.4 gives $\mathbf{Vec}(P_s, \bigoplus_t \mathbf{Vec}(Q_t, W)) \cong \bigoplus_t \mathbf{Vec}(P_s, \mathbf{Vec}(Q_t, W))$, and the tensor–hom adjunction (with P_s finite-dimensional) gives $\mathbf{Vec}(P_s, \mathbf{Vec}(Q_t, W)) \cong \mathbf{Vec}(P_s \otimes Q_t, W)$. Summing, $\cong \bigoplus_{(s,t)} \mathbf{Vec}(P_s \otimes Q_t, W)$. The unit: $(\{*\}, \mathbb{k}) \triangleleft (T, Q) = (\{*\} \times T, \mathbb{k} \otimes Q_t) = (T, Q)$. $\square$

Remark 7.2 (why the dependent sum disappears). Over $\mathbf{Set}$ the composite shapes are $\coprod_s T^{P_s}$: a composite shape is a shape s together with a function $P_s \rightarrow T$ choosing a T -shape at each position. Over $\mathbf{Vec}$, Lemma 4.4 says a linear map $P_s \rightarrow \bigoplus_t V_t$ is a *sum of components*, not a *choice of branch*; the “function $P_s \rightarrow T$ ” has nowhere to live. So the dependency T^{P_s} collapses to the plain product $S \times T$, and the positions tensor. Linearity flattens the dependent sum. This is the same $\coprod \subsetneq \bigoplus$ phenomenon once more, now inside composition.

Proposition 7.3 (comonoids: the algebroid guess, honestly). *A $\triangleleft$ -comonoid $((S, P), \delta, \varepsilon)$ in $(\mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}})_{\mathrm{fin}}, \triangleleft, I)$ amounts to: the shape set S (with its unique $(\mathbf{Set}, \times)$ -comonoid structure, the diagonal) together with, for each $s \in S$, a $\mathbb{k}$ -algebra structure on P_s . In particular a one-shape $\triangleleft$ -comonoid $(\{*\}, P)$ is exactly a $\mathbb{k}$ -algebra P .*

Proof (sketch; full verification in the companion note). The comultiplication $\delta: (S, P) \rightarrow (S, P) \triangleleft (S, P) = (S \times S, (P_a \otimes P_b))$ has shape component a function $S \rightarrow S \times S$; the counit $\varepsilon: (S, P) \rightarrow I = (\{*\}, \mathbb{k})$ has shape component $S \rightarrow \{*\}$. Coassociativity and counitality of the shape components make S a comonoid in $(\mathbf{Set}, \times)$, which forces the diagonal $s \mapsto (s, s)$ (every set is uniquely such a comonoid). With $a = b = s$, the position component of δ is a family $\mu_s: P_s \otimes P_s \rightarrow P_s$, and of ε a family $\eta_s: \mathbb{k} \rightarrow P_s$; coassociativity and counitality of δ become associativity and the unit laws for (μ_s, η_s) , i.e. a $\mathbb{k}$ -algebra on each P_s . Distinct shapes never interact, because the diagonal cannot reach the off-diagonal shape pairs (a, b) with $a \neq b$. $\square$

Remark 7.4 (the crown target, honestly assessed). The $\mathbf{Set}$ spine is poly-comonoid $\cong$ small category (directed containers $\cong$ categories); one hoped $\triangleleft$ -comonoid over $\mathbf{Vec} \cong$ $\mathbb{k}$ -linear category (algebroid, in Mitchell’s sense: a category enriched in $\mathbf{Vec}$). Proposition 7.3 shows the finite-dimensional analog is *weaker*: a $\triangleleft$ -comonoid yields only a *family of $\mathbb{k}$ -algebras* — a “diagonal” algebroid with no morphisms between distinct objects — because the linear composition $\triangleleft$ of Proposition 7.1 has lost the dependent-sum structure that, over $\mathbf{Set}$, lets a comonoid encode composable arrows between different objects. The biproduct/linearity collapse strikes a *third* time, now degrading algebroids to algebra-families. Recovering a full algebroid requires a genuinely different (lax, or bimodule) composition on $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$; pinning that down is future work, and is the true $\mathbf{Vec}$ analog of the equivalence $\mathbf{DCont} \cong \mathbf{Cat}$ — and §8 identifies that composition. **Status:** Proposition 7.3 is now *proved* (companion note 2026-08-19-vec-comonoids-algebras.md): the full statement is $\mathbf{Comon}_{\triangleleft}(\mathbf{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})) \cong \mathbf{Fam}(\mathbf{Alg}_{\mathbb{k}}^{\text{op}})$, with S *arbitrary* (finite S not needed), the comonoid morphisms being backward $\mathbb{k}$ -algebra homomorphisms, and an independent brute-force check over $\mathbb{k} = \mathbb{F}_2$ confirming that no off-diagonal comultiplication survives the laws.

8. PART 4 — THE CORRECT HOME: LINEAR DIRECTED CONTAINERS ARE $\mathbf{Vec}$ -MATRICES

Part 3 ended on an apparent defeat: the $\triangleleft$ -comonoids of $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ are only *families of algebras*, never a genuine algebroid, and we flagged that recovering the full Mitchell notion “requires a genuinely different (lax, or bimodule) composition.” This section identifies that composition. The punchline is not a new theorem — the mathematics here is classical, due to Bénabou and Mitchell — but a *diagnosis*: the naive single-index theory was working with the wrong shape of data, and once the data is corrected the algebroid reappears on the nose. The failure of Part 3 was not $\mathbf{Vec}$ ’s fault; it was a modelling error, and pinpointing it turns the collapse into a boundary we understand exactly.

8.1. The reframe: shapes are objects, positions are homs. Recall *why* $\mathbf{DCont} \cong \mathbf{Cat}$ works over $\mathbf{Set}$ (Ahman–Uustalu). In a directed container

the shapes are the *objects* of the category and the positions over a shape are the *arrows out of that object*; composition of the container is composition of arrows. The dictionary is *shapes* = *objects*, *positions* = *arrows*.

A single-index linear container $(S, (P_s))$ cannot honour this dictionary, and now we can see the reason sharply. It offers *one* position space P_s *per shape* — but a **Vec**-enriched category (an algebroid) has a hom-space $\text{hom}(a, b)$ for each *ordered pair* of objects. There is simply no room in the index $s \in S$ to record “an arrow *from a to b*.” In $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ the “shapes” behave as bare indices, not as objects with a homset between each pair. So the correct linear analogue of a directed container is not a family indexed by S but a family indexed by $S \times S$.

Definition 8.1 (*Vec-matrix / Vec-graph*). Fix a set S (the *objects*). A **Vec-matrix** on S is a family $P = (P_{a,b})_{(a,b) \in S \times S}$ of finite-dimensional $\mathbb{k}$ -vector spaces: one hom-space $P_{a,b}$ for each ordered pair. Write $\mathbf{Mat}_S(\mathbf{Vec})$ for the category of **Vec**-matrices on S and their entrywise linear maps. A **Vec-matrix** is exactly a **Vec**-enriched graph on the vertex set S ; equivalently it is a 1-cell $S \rightharpoonup S$ of the bicategory $\mathbf{Mat}(\mathbf{Vec})$ of sets, **Vec**-matrices, and matrix maps.

This is the expected directed-container reindexing made linear: shapes have become vertices/objects, and positions have become hom-objects *between* vertices — positions are arrows, and an arrow needs a source *and* a target.

8.2. The lax composition is matrix product. With the data corrected, the composition writes itself. Composing an arrow $a \rightarrow b$ with an arrow $b \rightarrow c$ should yield an arrow $a \rightarrow c$, summed over all possible intermediate objects b :

Definition 8.2 (*matrix composition*). For **Vec**-matrices P, Q on S , define $P \odot Q$ and the unit matrix I by

$$(P \odot Q)_{a,c} := \bigoplus_{b \in S} P_{a,b} \otimes Q_{b,c}, \quad I_{a,b} := \begin{cases} \mathbb{k} & a = b \\ 0 & a \neq b. \end{cases}$$

This is 1-cell composition in $\mathbf{Mat}(\mathbf{Vec})$ restricted to a fixed object set: matrix multiplication with $\otimes$ for entrywise product and $\bigoplus$ for entrywise sum. It makes $(\mathbf{Mat}_S(\mathbf{Vec}), \odot, I)$ a monoidal category.

Everything the collapse cost us is restored by the single symbol $\bigoplus_b$. Look at where it sits: the sum ranges over the *intermediate object* b , so it is the *composable-arrows sum* — the linear reincarnation of the **Set** dependent sum $\coprod_s T^{P_s}$ that flattened in Proposition 7.1. Over **Set** a composite shape is “a shape s together with a choice of next shape at each position”; that dependency is exactly a sum over the middle object, and the matrix product $\bigoplus_b P_{a,b} \otimes Q_{b,c}$ is its faithful linear image. One honesty caveat, stated once and meant precisely: this $\bigoplus_b$ is the *right reincarnation* of the extensivity coproduct whose absence ($\coprod \not\subseteq \bigoplus$) drove the whole collapse — it is a $\bigoplus$

over the middle object, deployed *where a categorical composite needs it* — but it is not literally the same coproduct as the one that failed to be disjoint in Part 1. The point is structural, not an identity of functors: putting the sum on the composition index, rather than on the shape index, is what a category wants and what a plain family cannot express.

8.3. Its (co)monoids are algebroids. Now the classical fact, in the two variances we need.

Proposition 8.3 (monoids in $\mathbf{Mat}_S(\mathbf{Vec})$ are algebroids; Bénabou). *A monoid (P, μ, η) in $(\mathbf{Mat}_S(\mathbf{Vec}), \odot, I)$ is exactly a $\mathbb{k}$ -linear category (algebroid, in Mitchell’s sense) with object set S : $\mathrm{hom}(a, b) = P_{a,b}$; the multiplication $\mu: P \odot P \rightarrow P$ is the family of composition maps $\bigoplus_b P_{a,b} \otimes P_{b,c} \rightarrow P_{a,c}$; and the unit $\eta: I \rightarrow P$ picks out the identity arrow $\mathbb{k} \rightarrow P_{a,a}$ at each object. Associativity of μ is associativity of composition; the unit laws are the identity-arrow laws. Thus $\mathbf{Mon}(\mathbf{Mat}_S(\mathbf{Vec}), \odot, I) \cong \mathbf{Vec}\text{-}\mathbf{Cat}_S$, the $\mathbb{k}$ -linear categories with object set S .*

This is Bénabou’s observation that a monad in a bicategory of matrices (or spans) is an enriched category, specialised to $V = \mathbf{Vec}$ over the discrete base S ; it is the additive avatar of the poly-comonad = category theorem (Spivak; the $\mathbf{Set}$ spine $\mathbf{DCont} \cong \mathbf{Cat}$). The *comonoid* version, which is what a container theory delivers after the contravariant twist on positions, follows by applying the fibrewise op .

Proposition 8.4 (comonoids, via the fibrewise op). *Work with $\mathbf{Vec}$ -matrices of finite-dimensional entries. The entrywise op $P_{a,b} \rightsquigarrow P_{a,b}^{\mathrm{op}}$ read in $\mathbf{Vec}^{\mathrm{op}}$ carries $\odot$ to $\odot$ (the tensor is self-dual on finite-dimensional spaces and $\bigoplus$ is its own dual), so a $\odot$ -comonoid in $\mathbf{Mat}_S(\mathbf{Vec}^{\mathrm{op}})$ is the same data as a $\odot$ -monoid in $\mathbf{Mat}_S(\mathbf{Vec})$ — both ops cancel. Hence*

$$\mathbf{Comon}(\mathbf{Mat}_S(\mathbf{Vec}^{\mathrm{op}}), \odot, I) \cong \mathbf{Vec}\text{-}\mathbf{Cat}_S$$

as well: $\mathbf{Vec}$ -enriched categories on object set S , now presented comultiplicatively. The variance bookkeeping (which op lands where) is the monad/comonad-in- V - $\mathbf{Mat}$ duality made explicit by Cruttwell–Shulman-style enriched double-category arguments (arXiv:1704.00329).

So the crown target is reached: on the corrected data — one hom-space per ordered pair — the linear $\triangleleft$ -comonoids *are* algebroids, exactly the $\mathbf{Vec}$ analogue of $\mathbf{DCont} \cong \mathbf{Cat}$. The open question 2 of Part 3 is answered in the affirmative; the “lax/bimodule composition” it asked for is the matrix product $\odot$.

8.4. The strict single-index $\triangleleft$ is the diagonal degeneration. Why, then, did Part 3 see only algebra-families? Because the single-index theory is the *diagonal* of the matrix theory — the arrows-only-between-equal-objects fragment.

Proposition 8.5 (diagonal embedding). *Send a single-index linear container $(S, (P_s))$ to the diagonal **Vec**-matrix $\widehat{P}$ with $\widehat{P}_{a,b} = P_a$ if $a = b$ and 0 otherwise. Diagonal **Vec**-matrices are closed under $\odot$:*

$$(\widehat{P} \odot \widehat{Q})_{a,c} = \bigoplus_b \widehat{P}_{a,b} \otimes \widehat{Q}_{b,c} = \begin{cases} P_a \otimes Q_a & a = c \\ 0 & a \neq c, \end{cases}$$

because the only surviving summand is $b = a = c$. So $\odot$ restricts, on diagonal matrices, to the pointwise tensor $(P_a \otimes Q_a)_a$ with unit $(\mathbb{k})_a$, and a $\odot$ -(co)monoid supported on the diagonal is precisely a family of $\mathbb{k}$ -(co)algebras — a family of one-object $\mathbb{k}$ -linear categories. This is the content of Proposition 7.3: $\text{Comon}_{\triangleleft}(\text{Fam}(\text{Vec}_{\text{fd}}^{\text{op}})) \cong \text{Fam}(\text{Alg}_{\mathbb{k}}^{\text{op}})$ is the diagonal fragment of $\text{Comon}(\text{Mat}_S(\text{Vec}^{\text{op}}), \odot, I) \cong \text{Vec-Cat}_S$.

The reconciliation with Part 3 is worth stating carefully, because the two compositions do differ off the diagonal. The strict single-index $\triangleleft$ of Proposition 7.1 has shape set the *full* product $S \times S$ and fibre $P_a \otimes P_b$ over every pair (a, b) — it does *not* agree with $\odot$, whose composite is a $\bigoplus_b$ and lives back on S . What Part 3 proved is that the counit *forces a single-index comonoid onto the diagonal* (§3 of the companion note: $\delta_{\text{shape}}(s) = (s, s)$), so the comultiplication only ever touches the fibres $P_s \otimes P_s \rightarrow P_s$ and the off-diagonal fibres $P_a \otimes P_b$ ($a \neq b$) are structurally unreachable. In other words the single-index comonoid theory *cannot escape the diagonal*, and on the diagonal it coincides with the genuine matrix theory of Proposition 8.5. The diagonalisation happens at two coupled levels that are really one phenomenon:

- (i) **position-support diagonal** — the embedding $\widehat{P}$ keeps only the $a = b$ hom-spaces (no arrows between distinct objects);
- (ii) **shape-comultiplication diagonal** — the counit forces $\delta_{\text{shape}} = \Delta$, so the comonoid comultiplication lands only in the (s, s) block.

Both say the same thing: *one object, no cross-object composition*, hence a $\mathbb{k}$ -algebra rather than a $\mathbb{k}$ -linear category. This is why the crown looked lost: the single-index framing had silently pre-restricted to the diagonal sub-bicategory before the classification even began.

8.5. The single obstruction, stated once. We can now name the one fact that runs through all four parts of this note. Over **Set** the coproduct is a disjoint union and the base is *extensive* (Carboni–Lack–Walters); over **Vec** the coproduct is a biproduct and extensivity fails, $\coprod \subsetneq \bigoplus$. Every collapse in this note is that one failure, viewed through a different structure:

- **objects** — shapes become invisible, a finite linear container is Id^N (Theorem 5.1);
- **morphisms** — the extension is not full, container morphisms are the $\coprod$ -indexed fragment of the $\bigoplus$ -indexed transformations (Theorem 6.3);

- **comonoids** — the $\triangleleft$ -comonoids of $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ are a family of algebras, the diagonal degeneration of an algebroid (Proposition 7.3, §8).

And the algebroid *reappears precisely when the matrix product restores the sum over the intermediate object*: the $\bigoplus_b$ of Definition 8.2 is the composition-index coproduct that the single-index theory had no place to put. The slogan of the whole note therefore sharpens from a diagnosis of failure into a statement of exactly where the boundary lies:

the naive linear $\mathbf{DCont} \cong \mathbf{Cat}$ fails because $\mathbf{Vec}$ is not extensive; it is repaired by moving the coproduct from the shape index to the composition index — from families over S to matrices over $S \times S$.

Provenance note (internal), §8. The mathematics of this section is *classical* and the section’s only contribution is the container-theoretic diagnosis (single-index = diagonal degeneration; the matrix $\bigoplus_b$ = the composition-index coproduct; non-extensivity as the single organising obstruction). *Monoid in a matrix/span bicategory = enriched category* is Bénabou (*Introduction to bicategories*, LNM 47, 1967); *algebroid = ring with several objects*, one object = algebra, is Mitchell (*Rings with several objects*, Adv. Math. 8, 1972); the monad/comonad-in- V - $\mathbf{Mat}$ variance guardrail (Proposition 8.4) is the enriched-duality-in-double-categories line, arXiv:1704.00329. The $\mathbf{Set}$ precedents lifted here are Ahman–Uustalu $\mathbf{DCont} \cong \mathbf{Cat}$ (arXiv:1604.01187) and Spivak’s polynomial-comonad = category (arXiv:2202.00534, 2111.10968); the non-extensivity of $\mathbf{Vec}$ is Carboni–Lack–Walters (JPAA 84, 1993). None of these is deep-read into **sources.json** this cycle; before any external circulation, direct-read D. Lin, *Enriched Polynomial Functors*, to confirm it does not already state this framing, and disambiguate from arXiv:1209.0940 / 1403.0833 (there “linear” means linear *logic*, not $\mathbf{Vec}$ -enrichment).

9. APPLICATION: HANDLING EITHER PROMPT IS THE BIPRODUCT

The note so far has read the biproduct as a *loss*: it erases shapes (§5) and breaks fullness (§6). This section turns the same fact over and reads it as a *gain*. The setting is the one that motivated the front in the first place — prompts and their responses — and the payoff is a clean categorical answer to a question Neil posed directly: *prompt A with reply-space A , prompt B with reply-space B ; what vector space matches being able to handle **either** prompt?* The answer is the coproduct of the two linear containers, and the biproduct makes that answer coincide with “be ready for **both**”.

9.1. Prompts as one-shape linear containers. Fix the modelling dictionary, which is Neil’s. A response to a prompt is not a single token but a *distribution* — a superposition of possible replies, carrying the model’s uncertainty — so the space of responses is a vector space, and the basis in which that space is expressed is what a learning system *learns*. Responses = uncertainty = the basis of learning. A prompt is therefore packaged as a

one-shape linear container: the single shape is the prompt, and its position space is the reply space.

Definition 9.1 (prompt container). A *prompt with reply-space* A is the one-shape linear container $\mathbf{A} := (\{*\}, A) \in \mathbf{Fam}(\mathbf{Vec}^{\text{op}})$, with extension the corepresentable

$$\llbracket \mathbf{A} \rrbracket W = \mathbf{Vec}(A, W) = h_A(W).$$

A point of $\llbracket \mathbf{A} \rrbracket W$ is a linear map $A \rightarrow W$: a way of reading the reply space into the ambient answer space W .

9.2. Either prompt is the coproduct; the coproduct is the biproduct. “Handle either prompt A or prompt B ” is a *choice of prompt*, and choice is coproduct. In $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ the coproduct is the disjoint union of shape sets — the one place a set-level disjoint union survives untouched, because it indexes shapes, not positions — so it is the honest “or”. The biproduct then does the rest.

Proposition 9.2 (either-prompt = biproduct). *Let $\mathbf{A} = (\{*\}, A)$ and $\mathbf{B} = (\{*\}, B)$ be prompt containers. Their coproduct in $\mathbf{LinCont} = \mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ is the two-shape container*

$$\mathbf{A} \sqcup \mathbf{B} = (\{*_A, *_B\}, (A, B)),$$

and its extension is the corepresentable at the biproduct $A \oplus B$:

$$\llbracket \mathbf{A} \sqcup \mathbf{B} \rrbracket = h_A \oplus h_B \cong h_{A \oplus B} = \llbracket (\{*\}, A \oplus B) \rrbracket.$$

Consequently, as functors, “handle either prompt A or prompt B ” is the single prompt whose reply-space is the biproduct $A \oplus B$.

Proof. The coproduct in $\mathbf{Fam}(\mathcal{C})$ is disjoint union of indexing sets with the position families juxtaposed, for any $\mathcal{C}$; hence $\mathbf{A} \sqcup \mathbf{B} = (\{*_A, *_B\}, (A, B))$. The extension functor sends this coproduct to the pointwise biproduct, $\llbracket \mathbf{A} \sqcup \mathbf{B} \rrbracket W = \mathbf{Vec}(A, W) \oplus \mathbf{Vec}(B, W) = (h_A \oplus h_B)(W)$, directly from Definition 3.3. Finally $h_A \oplus h_B \cong h_{A \oplus B}$ is the additivity of h (Lemma 4.3): h sends the biproduct $A \oplus B$ to the biproduct of corepresentables. The composite is $h_{A \oplus B} = \llbracket (\{*\}, A \oplus B) \rrbracket$. $\square$

Every step is an existing lemma; the content is entirely in what the identity *says*. Read it as a self-duality. The object $A \oplus B$ is a *biproduct*: it is simultaneously

$$\underbrace{A \oplus B}_{\text{coproduct: either } A \text{ or } B} = \underbrace{A \oplus B}_{\text{product: ready for both } A \text{ and } B}.$$

So the reply space that matches “I can answer either prompt” is the very same space that matches “I am prepared for both prompts at once”. Over **Set** these come apart: “answer either” is a tagged disjoint union $A \sqcup B$ (you must say *which* prompt you answered), while “ready for both” is the product $A \times B$; and $A \sqcup B \neq A \times B$. The collapse of the two is exactly what **Vec** buys and **Set** forbids.

Remark 9.3 (Neil’s intuition is the biproduct). The design intuition behind the question — *coproducts should survive, but also carry the universal property of products, so that passing to “either” loses nothing* — is not a wish to be arranged for; it is a *theorem*, and its name is the biproduct. “Loses nothing” is precisely the statement that the coproduct injections $A, B \hookrightarrow A \oplus B$ are complemented by projections $A \oplus B \rightarrow A, B$, i.e. that the coproduct *is* the product. In $\mathbf{Vec}$ this holds for free (§2); demanding it of $\mathbf{Set}$ would force the summands to overlap, which a disjoint union refuses. The either-prompt reading is thus the cleanest possible statement of why one might *want* to work over a non-extensive, additive base: it is the base on which “or” automatically supplies “and”.

9.3. The collapse, re-read as a feature. Proposition 9.2 is the biproduct collapse of §5 seen from the other side. There the additivity of h was a *liability*: it erased the shape set, so that $(\{*\}, A \oplus B)$ and $(\{*_A, *_B\}, (A, B))$ — different containers — have the same extension, and no representation theorem can tell them apart (Theorem 5.1, Corollary 6.4). Here the identical mechanism is an *asset*: it is exactly what lets “either” inherit the universal property of “both”. The collapse and the application are one equation $h_A \oplus h_B \cong h_{A \oplus B}$ read with opposite valuations.

The relation is global-versus-local. The collapse theorem $\llbracket S, P \rrbracket \cong \text{Id}^N$ is the global biproduct fact — *all* finite shape structure flattens to a single multiplicity N . Proposition 9.2 is its local face on a single binary coproduct: two prompts merge into one, their reply spaces into the biproduct. Same additivity of h ; whether it reads as loss or gain is a matter of what one is trying to do. For a representation theorem, losing the shapes is a defect. For an agent that must answer whatever it is asked, merging the reply spaces into a single space that is at once “either” and “both” is exactly right.

9.4. Composing responses: within a prompt and across prompts. The either-prompt reading also draws the line between the two composition structures the note has isolated, and shows the biproduct doing double duty. A response can be composed with a further response in two genuinely different ways.

Within a single prompt. Chaining a response of prompt A into another response of the *same* prompt is an operation $A \otimes A \rightarrow A$ on one reply space. This is the $\mathbb{k}$ -algebra structure that a single-shape $\triangleleft$ -comonoid carries: by Proposition 7.3 a one-shape linear directed container is exactly a $\mathbb{k}$ -algebra (A, μ, η) , with $\mu: A \otimes A \rightarrow A$ the within-prompt composition and $\eta: \mathbb{k} \rightarrow A$ the empty response. No sum over intermediates appears: there is only one prompt, so there is nothing to sum over.

Across distinct prompts. Chaining a response of prompt a into a response of a *different* prompt c , routed through some intermediate prompt b , is the matrix product of §8:

$$(P \odot Q)_{a,c} = \bigoplus_b P_{a,b} \otimes Q_{b,c}.$$

Here the reply data is a **Vec**-matrix — one hom-space $P_{a,b}$ *per ordered pair of prompts* — and composition sums over every intermediate prompt b . The $\bigoplus_b$ is the composable-responses sum, and it is available for exactly the reason of this section: **Vec** has biproducts. The comonoids of this composition are algebroids (Proposition 8.4), and the within-prompt algebras are its diagonal degeneration (Proposition 8.5).

So the biproduct that makes “either” equal “both” (Proposition 9.2) is the same biproduct that supplies the $\bigoplus_b$ powering cross-prompt composition. One structural fact — **Vec**’s finite coproduct is a biproduct — underwrites both the either/both self-duality on a single coproduct and the composition of responses across a whole network of prompts. The single-index theory sees only the diagonal: responses composing within one prompt. The matrix theory restores the off-diagonal: responses composing between prompts. Moving from “one prompt, an algebra” to “many prompts, an algebroid” is moving the biproduct from inside a reply space to between reply spaces — the same relocation of the coproduct, from the shape index to the composition index, that §8 identified as the repair of $\mathbf{DCont} \cong \mathbf{Cat}$.

Provenance note (internal), §9. The mathematics is elementary and classical: the biproduct (finite coproduct = product in an additive category) is standard; the coproduct in **Fam** is disjoint union of indices; Proposition 9.2 is Lemma 4.3 applied to a two-element family. The contribution is the *reading*: prompts as one-shape linear containers, “handle either” as the container coproduct, and the identification of that coproduct’s reply space with the biproduct $A \oplus B$ — so that “either” carries the universal property of “both”. The prompt/response modelling and the “responses = uncertainty = basis of learning” motivation are Neil’s (2026-08-21 correspondence); the container framing and the either=both diagnosis are the note’s. Non-extensivity of **Vec** is Carboni–Lack–Walters (JPAA 84, 1993); the algebroid facts of §9.4 are Bénabou/Mitchell, as attributed there.

10. THE CARRY-OVER TABLE

The following table is the honest scorecard of the base change. “Proved” means proved in the companion note; “computed” means checked in examples and on the main components but not yet fully verified; “open” means genuinely open.

Set fact	Vec analog	status
$\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Set}^{\mathrm{op}})$, extension $\sum_s X^{P_s}$	$\mathbf{LinCont} := \mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}})$, extension $\bigoplus_s \mathbf{Vec}(P_s, -)$	def. (3.3)
$F(1) = S$ recovers shapes	fails: $F(0) = 0$; shapes = indecomposable summands	proved (5.1)
$\llbracket - \rrbracket$ fully faithful (rep. theorem)	full only up to $\coprod \subsetneq \bigoplus$; finite recovery only of N	proved (6.3, 6.4)
composition = dependent sum $\coprod_s T^{P_s}$	$(S \times T, P_s \otimes Q_t)$: dependency flat-tens	proved (7.1)
poly-comonoid $\cong$ small category	$\triangleleft$ -comonoid in $\mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}})$ = family of $\mathbb{k}$ -algebras (the <i>diagonal</i> degeneration of an algebroid)	proved (7.3)
$\mathbf{DCont} \cong \mathbf{Cat}$ (directed containers)	$\mathbf{Comon}(\mathbf{Mat}_S(\mathbf{Vec}^{\mathrm{op}}), \odot, I) \cong \mathbf{Vec}\text{-}\mathbf{Cat}_S$: matrix product recovers the full algebroid	classical (§8)
“either A or B ” = tagged union $A \sqcup B \neq A \times B$	coproduct of prompt containers has reply-space the biproduct $A \oplus B$: “either” = “both”	proved (9.2)
Day-convolution tensors $\leftrightarrow$ monoidal structures on $\mathbf{Set}$	convolutional tensors on open $\mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}}) \leftrightarrow$ monoidal structures on $\mathbf{Vec} (\otimes_{\mathbb{k}}, \oplus)$	
general rep. theorem (which F are containers)	which $F: \mathbf{Vec} \rightarrow \mathbf{Vec}$ are small co-products of corepresentables	open (§13)

Every “fails” / “only up to” row in this table is one avatar of $\coprod \subsetneq \bigoplus$. Reading the table top to bottom is reading the single non-extensivity of $\mathbf{Vec}$ propagate through the theory.

11. THE ANALYTIC LIFTING: SCHUR FUNCTORS AND THE DEGREE-ONE CORNER

There is a second, well-established way to lift the word “container” to $\mathbf{Vec}$, and a reader who knows representation theory will have it in mind: not the additive, hom-based lifting of this note but the *analytic*, tensor-based one — Schur functors and vector species. One should ask whether that theory already classifies our objects, or even scoops the programme. It does neither. Setting the two liftings side by side locates ours precisely: it is the *degree-one corner* of the analytic lifting, the corner where the whole Young-diagram classification degenerates to the biproduct collapse of §5. We work in characteristic 0 throughout this section (Remark 11.3 flags what breaks otherwise).

11.1. Two liftings, side by side. The *additive (hom-based) lifting* is ours: a linear container $(S, (P_s))$ has extension $\llbracket S, P \rrbracket W = \bigoplus_{s \in S} \mathbf{Vec}(P_s, W)$, with

positions in the *hom slot* $\mathbf{Vec}(P_s, -)$. It is a $\mathbf{Vec}$ -functor — $\mathbb{k}$ -linear on hom-spaces — and, being additive, is *homogeneous of degree 1*: under $W \xrightarrow{\lambda} W$ its value scales by λ .

The *analytic (tensor-based) lifting* starts instead from a *vector species*: a sequence $M = (M_n)_{n \geq 0}$ in which M_n is a representation of the symmetric group $\mathfrak{S}_n$ (equivalently a functor $\mathbf{core}(\mathbf{FinSet}) \rightarrow \mathbf{Vec}$). Its *analytic functor* is

$$W \mapsto \bigoplus_{n \geq 0} (M_n \otimes W^{\otimes n})_{\mathfrak{S}_n},$$

with positions now in the *tensor powers* $W^{\otimes n}$. In characteristic 0 this functor category is semisimple, and its simple objects are the *Schur functors* $\mathbb{S}_\lambda$, indexed by Young diagrams λ ; each $\mathbb{S}_\lambda$ is homogeneous of degree $|\lambda|$ (the number of boxes), and every polynomial functor decomposes as $\bigoplus_\lambda M_\lambda \otimes \mathbb{S}_\lambda$ with M_λ a multiplicity space. The smallest cases are $\mathbb{S}_{(1)} = \text{Id}$, $\mathbb{S}_{(n)} = \text{Sym}^n$, and $\mathbb{S}_{(1^n)} = \Lambda^n$.

The structural fork, in one line. A Schur functor acts on the scalar $\lambda \cdot \text{id}_W$ by $\lambda^{|\lambda|} \cdot \text{id}$. Hence

$$\mathbb{S}_\lambda \text{ is } \mathbb{k}\text{-linear on hom-spaces} \iff |\lambda| = 1.$$

The analytic lifting leaves the enriched ($\mathbb{k}$ -linear-on-homs) world the moment its degree exceeds 1; the additive lifting never leaves it. Above degree 1 the two liftings do not merely differ — they inhabit different functor categories.

11.2. The degree-one corner. The one place the two liftings meet is degree 1, and there they agree exactly.

Proposition 11.1 (the additive lifting is the degree-one slice). *The only Schur functor of degree 1 is $\mathbb{S}_{(1)} = \text{Id}$. Consequently the degree-1 homogeneous part of any polynomial functor is $M_{(1)} \otimes \text{Id} \cong \text{Id}^N$ with $N = \dim M_{(1)}$ — which is exactly the class of finite linear container extensions (Theorem 5.1) — and the multiplicity space $M_{(1)}$ is the single number N that survives the collapse.*

Proof. $\deg \mathbb{S}_\lambda = |\lambda|$, and $|\lambda| = 1$ forces $\lambda = (1)$, so $\mathbb{S}_{(1)} = \text{Id}$ is the only degree-1 Schur functor. The degree-1 part of $\bigoplus_\lambda M_\lambda \otimes \mathbb{S}_\lambda$ is therefore $M_{(1)} \otimes \text{Id} \cong \text{Id}^{\dim M_{(1)}}$. By Theorem 5.1 a finite linear container has extension Id^N ; the two match with $N = \dim M_{(1)}$. $\square$

The match is exact, and it is a *degeneration*. In the full semisimple category the Young-diagram bookkeeping records an entire family of shapes $(\mathbb{S}_\lambda)_\lambda$ with multiplicities $(M_\lambda)_\lambda$; in the degree-1 corner there is a *single* indecomposable, $\mathbb{S}_{(1)} = \text{Id}$ (compare Remark 5.2, whose cocontinuous Id^N is the same class), and the bookkeeping has nothing left to say beyond counting its multiplicity N . That degeneration is the biproduct collapse, seen from the representation-theoretic side.

Degree 2 is where the analytic lifting produces its first genuinely new functors, and none of them is a container.

Example 11.2 (degree 1 versus degree 2 on $W = \mathbb{k}^2$). Write $W = \mathbb{k}^2$, so $W^{\otimes 2} = \mathbb{k}^4$.

functor	value on $\mathbb{k}^2$	dim	a linear container?
$\mathrm{Id} = \mathbb{S}_{(1)}$	$\mathbb{k}^2$	2	yes : $(\{*\}, \mathbb{k})$
Id^N	$(\mathbb{k}^2)^N$	$2N$	yes : any (S, P) , $\sum_s n_s = N$
$\mathrm{Sym}^2 = \mathbb{S}_{(2)}$	$\mathrm{Sym}^2 \mathbb{k}^2$	3	no — degree 2
$\Lambda^2 = \mathbb{S}_{(1,1)}$	$\Lambda^2 \mathbb{k}^2$	1	no — degree 2
$W^{\otimes 2} = \mathbb{S}_{(2)} \oplus \mathbb{S}_{(1,1)}$	$\mathbb{k}^4$	4	no — degree 2

The obstruction is additivity. A linear container satisfies $\llbracket S, P \rrbracket(V \oplus W) = \llbracket S, P \rrbracket V \oplus \llbracket S, P \rrbracket W$, whereas

$$\mathrm{Sym}^2(V \oplus W) = \mathrm{Sym}^2 V \oplus (V \otimes W) \oplus \mathrm{Sym}^2 W,$$

and the cross term $V \otimes W$ is exactly what an additive functor may not have. So Sym^2 and Λ^2 are not additive — indeed $\mathrm{Sym}^2(\lambda \cdot \mathrm{id}) = \lambda^2 \cdot \mathrm{id}$, so they are not even $\mathbf{Vec}$ -functors — and neither is the extension of any linear container.

11.3. The verdict, and what Schur does not touch. The relationship is neither coincidence nor irrelevance. It is not an exact coincidence: the classification statements differ — ours has a single indecomposable Id , the Schur category has all of the $\mathbb{S}_\lambda$. It is not “no relation”: the additive lifting is *literally the degree-one slice* $M_{(1)} \otimes \mathrm{Id}$ of the analytic one, and the honest replacement slogan of §5 — “shapes = indecomposable summands” — is precisely the Krull–Schmidt principle that, in the full semisimple category, recovers the $\mathbb{S}_\lambda$ and their multiplicities. The meta-principle is shared; only its instantiation degenerates, to “one shape Id , multiplicity N ”.

What the Schur/species picture does *not* reach is exactly the content of Parts 2 and 3 of this note.

The morphism layer.: Schur–species theory is a statement about the analytic functor category itself; it never forms the comparison $\llbracket - \rrbracket : \mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}}) \rightarrow [\mathbf{Vec}, \mathbf{Vec}]$, and so never sees that this functor is faithful but not full. The extensivity crux $\coprod \subsetneq \bigoplus$ (Theorem 6.3) is invisible to it.

The $\triangleleft$ -comonoid axis.: Species do carry a substitution product (plethysm), the analytic analogue of $\triangleleft$, and our $\triangleleft$ restricted to extensions is its degree-one shadow ($\mathrm{Id}^N \circ \mathrm{Id}^M = \mathrm{Id}^{NM}$). But the comonoid classification *in the shape-indexed category* $(\mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}}), \triangleleft)$ — and the finding (Proposition 7.3) that it degrades to families of $\mathbb{k}$ -algebras rather than reaching Mitchell’s algebroids — is a statement about the shape-indexed presentation, which species theory does not organise.

So the honest one-sentence placement, for the grant’s theory pillar: the characteristic-0 theorem “Schur functors = polynomial species” classifies the *analytic* lifting of containers to **Vec** and does not scoop the linear-container programme — the additive lifting is its degree-one corner, where the Young-diagram classification collapses to Id^N , and the programme’s real content (the extensivity crux on morphisms, the $\triangleleft$ -comonoid axis) lives outside the Schur picture entirely.

Remark 11.3 (the characteristic- p trap). Everything above is characteristic 0, the regime in which “Schur functors = polynomial species” and the semisimple $\bigoplus_\lambda \mathbb{S}_\lambda$ decomposition hold. Over characteristic p they fail — already in characteristic 2, $\mathbb{S}_{(2)}$ and $\mathbb{S}_{(1,1)}$ agree pointwise yet differ as functors — so the “degree-one corner” reading must not be exported to characteristic p without redoing the comparison. The biproduct collapse itself (§5) is characteristic-independent: $\llbracket S, P \rrbracket W = W^N$ over any field.

12. THE NEIGHBOURS, HONESTLY

This section is the prior-art ledger. The container framing is an *assembly*; the pieces are owned elsewhere, and it matters to say so precisely.

Strict polynomial functors.: Friedlander–Suslin introduced strict polynomial functors on **Vec** (built from $\otimes$, symmetric powers Sym , exterior powers Λ , direct sums), organised by degree d into categories $\mathcal{P}_d$; Krause and Touzé develop the theory extensively. Our objects $\bigoplus_s h_{P_s}$ are exactly the *additive* (= degree-one) corner, $\mathcal{P}_1 \simeq \mathbf{Vec}$, whose corepresentables are $\mathbf{Vec}(P, -) \cong (-)^{\dim P}$. *This literature owns the objects.* What it does *not* do is index a *shape set* over its additive part — and Theorem 6.3 explains why it never needed to: the shape set is not an invariant of the functor. The container framing is what introduces the shape index; the biproduct collapse is the price of doing so over a non-extensive base. §11 works this comparison out in full: our additive lifting is the degree-one corner of the Schur/analytic lifting, and the slogan “shapes = indecomposable summands” is the Krull–Schmidt principle that, in higher degree, recovers the Schur functors $\mathbb{S}_\lambda$ — but in the additive corner has only Id to find, so it degenerates to counting N . That meta-principle is thus partially anticipated in spirit; the container-side novelty is the assembly below, not the principle.

Vector species / twisted commutative algebras.: Species valued in **Vec** (Joyal’s species linearised; Sam–Snowden’s twisted commutative algebras) carry the analytic/Day-convolution structure — the “ $\bigoplus_n V_n \otimes_{\Sigma_n} (-)^{\otimes n}$ ” expansion, whose simple pieces in characteristic 0 are the Schur functors. That is the analytic lifting dissected in §11 and the natural home of the open Day-tensor row of the table; its substitution product (plethysm) is the analytic $\triangleleft$, of which our

composition (Proposition 7.1) is the degree-one shadow. It is all theirs.

Familial representability and extensivity.: “Coproduct of representables” with the disjointness (extensivity) hypothesis that makes the representation theorem work is Diers’ familial/multi-representability, and Carboni–Johnstone’s analysis of when such functors are well-behaved. Our Theorem 6.3 is, in this language, the observation that dropping extensivity is *exactly* what breaks fullness — a base-change reading of a hypothesis they already isolated.

Algebroids.: “ $\mathbb{k}$ -linear category” = “ring with several objects” is Mitchell’s; a one-object algebroid is a $\mathbb{k}$ -algebra. That is the target of the comonoid row, and Proposition 7.3 reports honestly that the naive single-index $\triangleleft$ reaches only the diagonal (a family of algebras). §8 then reaches the full Mitchell notion, on the correct data: it is Bénabou’s theorem that a monoid in the matrix bicategory $\mathbf{Mat}(\mathbf{Vec})$ is a $\mathbf{Vec}$ -enriched category, of which the single-index theory is the diagonal fragment. That identification is classical; the container-side novelty is only the diagnosis that single-index = diagonal.

The classical tools.: Krull–Schmidt–Azumaya (unique decomposition into indecomposables over a semiperfect endomorphism ring) and the Eilenberg–Watts theorem ($\mathbf{Vec}$ -cocontinuous endofunctors are $V \otimes -$) are standard; we use them as named theorems.

The delta, stated plainly. The contribution of the container view is not any object or any classical theorem above. It is the *assembly*: the $\mathbf{Fam}(\mathbf{Vec}^{\mathrm{op}})$ framing, the shape index it puts on top of degree-one strict polynomial functors, and — the load-bearing observation — the identification of the shape-collapse (on objects, Theorem 5.1) and the fullness gap (on morphisms, Theorem 6.3) as *one* phenomenon: the failure of extensivity $\coprod \subsetneq \bigoplus$ under $\mathbf{Set} \rightsquigarrow \mathbf{Vec}$.

Provenance note (internal). Every attribution in this note — the neighbour references here (Friedlander–Suslin, Krause, Touzé, Sam–Snowden, Diers, Carboni–Johnstone, Mitchell, Eilenberg–Watts, Krull–Schmidt–Azumaya) *and* the representation-theoretic facts of §11 (Schur–Weyl; the characteristic-0 identification “Schur functors = polynomial species”; the semisimple $\bigoplus_{\lambda} \mathbb{S}_{\lambda}$ decomposition and the degree bookkeeping $\deg \mathbb{S}_{\lambda} = |\lambda|$) — rests on general mathematical knowledge and has *not* been deep-read into `sources.json` this cycle. None of these citations is load-bearing: the mathematical content stands on the proofs of §§5–7 and the elementary degree calculation of §11, not on the attributions. Before any external circulation, each must be verified against the primary source in a browse session; do not upgrade any of them to a load-bearing citation without that check.

13. CONNECTIONS AND OPEN QUESTIONS

The picture. There is one organising slogan and everything hangs off it:

$$\boxed{\coprod \subsetneq \bigoplus} \quad (\mathbf{Set}: \text{equal, extensive} \implies \mathbf{Vec}: \text{strict, non-extensive}).$$

Objects: $\coprod$ keeps shapes apart, $\bigoplus$ merges them $\Rightarrow$ finite collapse (5.1).

Morphisms: $\coprod$ is one branch, $\bigoplus$ is a combination $\Rightarrow$ non-fullness (6.3).

Composition: $\coprod_s T^{P_s}$ (dependent sum) flattens to $S \times T$ (7.1). Comonoids: diagonal only $\Rightarrow$ family of algebras, not algebroid (7.3).

Open questions.

- (1) **General representation theorem.** For arbitrary (infinite S , infinite-dimensional P_s) linear containers, characterise the essential image of $\llbracket - \rrbracket$ in $[\mathbf{Vec}, \mathbf{Vec}]$ intrinsically. Candidate: accessible functors that are small coproducts of corepresentables. A positive answer would give an equivalence onto a describable subcategory, up to the biproduct ambiguity of Corollary 6.4, presumably via a Krull–Schmidt / generic-factorization statement in $[\mathbf{Vec}, \mathbf{Vec}]$; the obstruction to a clean equivalence is exactly Diers’ extensivity hypothesis, which $\mathbf{Vec}$ fails. Open beyond the finite case.
- (2) **The right composition for algebroids — resolved, then sharpened.** §8 answers the original question: the matrix product $\odot$ on $\mathbf{Vec}$ -matrices $\mathbf{Mat}_S(\mathbf{Vec})$ has comonoids = genuine $\mathbb{k}$ -linear categories (Proposition 8.4), and the single-index $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$ theory is its diagonal degeneration. The classical part being settled, the live questions become: (a) is $\mathbf{Mat}(\mathbf{Vec})$ the extension of a *container-style* monoidal structure on a category of “linear directed containers” with a clean extension functor into $[\mathbf{Vec}, \mathbf{Vec}]$, or only a bicategory presentation? (b) does the $\odot$ -comonoid = algebroid equivalence survive dropping finite-dimensionality of the hom-entries, where the self-duality used in Proposition 8.4 fails? (c) is the diagonal embedding (Proposition 8.5) the *unique* way the single-index theory sits inside the matrix theory?
- (3) **Convolutional tensors.** Classify the closed convolutional tensors on $\mathbf{Fam}(\mathbf{Vec}^{\text{op}})$; conjecturally they correspond to monoidal structures on $\mathbf{Vec}$ ($\otimes_{\mathbb{k}}$ and $\oplus$), the $\mathbf{Vec}$ -analog of the author’s $\mathbf{Set}$ classification. This is the Day-family / vector-species row of the table.
- (4) **Infinite-dimensional composition.** Proposition 7.1 uses finite-dimensional positions (tensor–hom). Describe $\triangleleft$ when positions are infinite-dimensional, where h_P becomes a product functor (Lemma 4.4).

Why this matters for the grant. The equivalence-chain spine $\mathbf{Containers} \simeq \mathbf{Dir.Cont} \simeq \mathbf{Poly-comonoids} \simeq \mathbf{Cat}$ is the theoretical heart of the AI Mathematician programme. This note is the first honest measurement of how that spine deforms under a base change to an abelian category: two of its four

joints (shape-recovery, full faithfulness) snap at the same rivet (extensivity), one (composition) survives in altered form, and the last (comonoids = categories) first appears to degrade to comonoids = algebra-families — until §8 shows this is a *modelling* artefact: on the correct data, one hom-space per ordered pair of objects, the linear comonoids are exactly algebroids ($\mathbf{Vec}$ -enriched categories), and the family-of-algebras answer is the diagonal degeneration. The value is diagnostic: it tells us that a representation-theoretic version of the programme is possible, that it must route around non-extensivity on objects and morphisms, but that the categorical spine itself (directed containers = categories) lifts cleanly once the coproduct is moved from the shape index to the composition index — and it names the one obstruction, extensivity, that every part of the story turns on. For the grant’s applied half the same measurement pays a dividend: the biproduct that obstructs the representation theorem is exactly what an agent wants. Modelling a prompt as a one-shape linear container (§9), “handle either prompt” is the container coproduct, and its reply-space is the biproduct $A \oplus B$ — so “answer either” and “be ready for both” are one object, and the very same $\oplus$ supplies the composable-responses sum of the algebroid. Non-extensivity is a defect for a classification and a feature for a compositional model of prompt/response systems; which face one sees is a matter of what one is building.

Companion: the proofs of Theorems 5.1, 6.1–6.3, Corollary 6.4, and Proposition 7.1 are in `proofs/2026-08-18-linear-containers-vec.md`; the full comonoid classification (Proposition 7.3) is proved in `proofs/2026-08-19-vec-comonoids-algebras.md`, with computational checks in `scratch/vec-containers/verify.py` and `scratch/vec-comonoids/verify.py`. The matrix-bicategory framing of §8 is classical (Bénabou, Mitchell); this note only records the container-theoretic diagnosis.